

Article

Techno-Economic Design and Optimization of Hybrid Energy Systems

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Abstract: The power generation capacity must be increased to accommodate population growth and address the lack of electricity access in rural areas. Traditional power plants in South Africa are unable to keep up with the growing demand for electricity. By strategically planning and building clusters of renewable energy sources like solar and wind, microgrid operators can provide a sustainable solution that boosts electricity supply while being cost-effective and environmentally friendly. Utilizing renewable energy can help alleviate strain on power plants by reducing peak demand in constrained distribution networks. The benefits of renewable energy include lower electricity expenses, enhanced system reliability, investment reallocation, and reduced environmental impact. These advantages will enhance the efficiency of the power system and contribute economic value to society. However, integrating solar power into the network infrastructure presents challenges such as fundamental changes in network structure, its intermittent nature due to unpredictability, and geographical constraints, which can complicate the task of grid operators in balancing electricity supply and demand within system limits while minimizing costs. The study employs Homer Pro 3.18.1 software to assess the economic costs and benefits of effectively integrating renewable technologies into the power grid. The aim is to evaluate the economic and technical feasibility of investing in renewable energy projects within the network. The research outcomes can guide power system operators, planners, and designers in leveraging solar energy to drive economic growth and industrial advancement, as well as assist independent power producers in making informed investment choices.



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Keywords: cost of energy; electricity network; hybrid energy optimization; reduced electricity prices

1. Introduction

How people in rural areas adapt during energy crises is a serious subject that cannot be disregarded. Energy poverty is prevalent in rural areas. In this study, energy poverty is defined as an insufficient quantity of energy resources to meet the related human energy demand. In terms of cost, energy poverty is defined as a condition in which people cannot afford to use modern, somewhat more expensive renewable energy sources [1]. Energy poverty in rural places does not occur in isolation. It is a manifestation of current income, social, and access disparities. For example, energy poverty primarily affects low-income households, which spend a large amount of their income on energy. Approximately 75% of South African households are without power [2]. Currently, around 80% of South Africa's rural population has access to electricity. However, rural low-income households are having a difficult time meeting their energy needs. Non-electrified households rely on filthy energy sources. Some households with electricity, particularly low-income ones, rely on government-provided electricity because they cannot purchase it. When that energy is depleted, poor households turn to less environmentally friendly sources of energy, such as firewood. Even though 80% of rural residents are linked to the grid, low-income households continue to face barriers to renewable energy access because they rely on more

unclean, unhealthy, and dangerous energy sources such as paraffin, firewood, and cow dung [3]. There is substantial inequity regarding renewable energy access in rural areas. Access to sustainable modern energy technologies is critical for promoting health, quality education, and long-term livelihoods; however, millions of people lack access to sustainable energy technologies, forcing them to rely on dangerous and unhealthy fuels for lighting and cooking. Despite the availability of many energy technologies in South Africa, rural households continue to struggle to meet their energy needs. For several years, certain rural regions have struggled with insufficient national grid coverage and poor electrical supplies. The government has made numerous efforts to connect grids in various parts of the country, including the Eastern Cape and KwaZulu Natal. However, even in electrified areas, people are unable to pay due to poverty, indicating that energy poverty renders access to power unpredictable. Rural areas in KwaZulu Natal are among the regions with the lowest access to renewable energy, despite huge energy resources like as wind, solar, biomass, and micro-hydrokinetic energy.

The increasing relevance of sustainable and affordable energy solutions has led to the optimization of grid connections with renewable energy for low-income families. As the globe faces the difficulties of climate change and rising energy costs, it is essential to investigate alternate energy sources that can help underprivileged areas. Low-income households frequently bear the weight of high energy costs and are more heavily impacted by environmental degradation. As a result, determining how to optimize grid connection with renewable energy can lead to considerable improvements in these households' livelihoods.

In South Africa, coal-fired facilities provide approximately 78% of the power [4], the majority of which were built during the apartheid era in the 1960s and 1970s. As a result of utilizing coal, South Africa ranks seventh in terms of per capita carbon dioxide emissions globally and accounts for approximately 40% of carbon dioxide emissions in Africa [5]. Load shedding is currently a problem in South Africa due to aging infrastructure and corruption in coal purchasing and power plant procurement, which has resulted to Eskom's (South African utility supply) USD 24 billion debt. Addressing the growing demand for power is a major challenge due to the high cost and greenhouse gas emissions associated with traditional energy sources. Increasing the use of renewable energy systems can assist to reduce carbon emissions caused by greenhouse gases. There are several problems with electric power generation, transmission of power from the generating station to the user, and subsequently distribution in rural areas because of the increasing prices. How people in rural areas adjust during energy crises is a crucial issue that should not be overlooked. Energy poverty is common in rural areas, characterized as a lack of energy supplies to meet the associated human energy demand. In terms of cost, energy poverty refers to a situation in which people are unable to acquire electricity from traditional or modern sources. The percentage of South African houses connected to the grid increased from 76.7% in 2002 to 89.6% by 2022. Figure 1 shows the percentage distribution of households that are connected to Eskom Grid [6]. However, rural low-income households are having a difficult time meeting their energy needs. Non-electrified households rely on filthy energy sources. Figure 2 shows the percentage distribution of the main sources of energy used for cooking by province in 2022, extracted in [6]. The figure illustrates that the percentage of households using electricity for cooking increased from 57.5% in 2002 to 76.5% in 2022. This increase was matched by an increase in the percentage of families using alternate sources of electricity, such as generators. This type of energy for cooking increased from 1.2% in 2014 to 7.8% in 2019, before declining to 4.8% in 2022. This form of energy for cooking rose from 1.2% in 2014 to 7.8% in 2019, before falling to 4.8% in 2022. The percentage of houses that used gas (mostly standard Liquefied Petroleum Gas) also rose from 2.2% in 2002 to 6.7% in 2022. Since 2002, there has been a significant drop in the use of paraffin, coal, and firewood. The percentage of homes that used paraffin collapsed from 16.1% in 2002 to 2.8% in 2022, while the percentage of households that used firewood decreased from 20% to 7.7%. The number of families who uses electricity as their primary source of energy for cooking is the highest in the Free State (86.2%) and KwaZulu-Natal (82.0%), and

lowest in Limpopo (63.5%). Other sources of electricity (such as generators) are the most common in Gauteng (9.0%) and Northwest (6.8%). Paraffin is the most used in Gauteng (5.3%), with the Western Cape having the lowest utilization (0.2%). The usage of wood and coal remained most notable in Limpopo (34.0%), Mpumalanga (18.0%), the Eastern Cape (9.3%), the North-West (8.4%), and KwaZulu-Natal (8.2%). In the Western Cape and Gauteng, less than 1% of families utilized wood to cook (0.5% and 0.7%, respectively). Gas is the most used by households in the Western Cape (19.9%) and Northern Cape (12.0%). South Africa's power network consists primarily of major power facilities located inland near Gauteng's mining cities and industrial sectors. They have an extensive transmission infrastructure to reach the coastline.

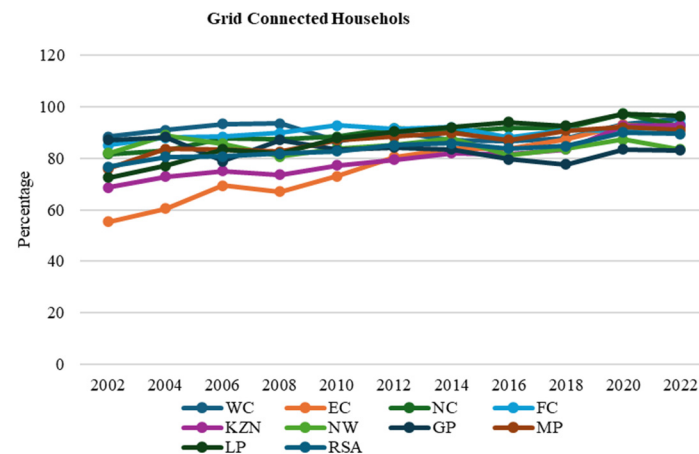


Figure 1. Grid Connected household percentage distribution by Province.

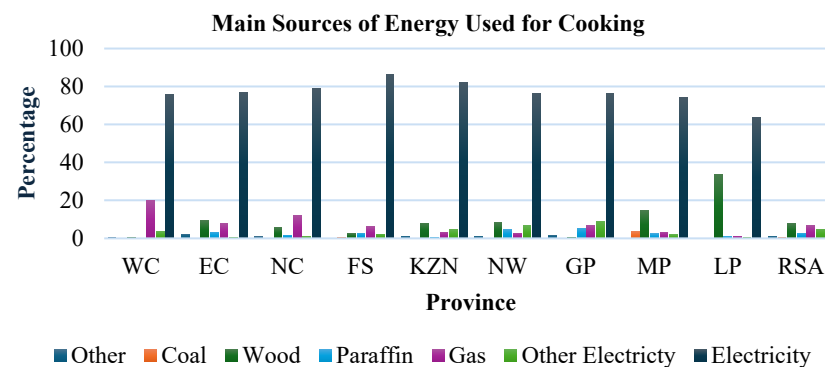


Figure 2. Main sources of energy used for cooking by province, 2022.

It is vital to note that Eskom's local coal activities provide more than 90% of South Africa's electricity. Municipalities purchase large quantities from the utility and distribute them to ratepayers. The mining and manufacturing industries buy directly from Eskom, accounting for 40% of the company's distribution activity [7]. From the previous studies, researchers highlighted the following eight major difficulties to electricity supply:

- Electricity pricing reflection and a lack of accessibility in subsidies programs.
- Poor municipal electricity distribution, which could worsen and result in significant economic expense.
- Eskom has inadequate management, low plant reliability, and ineffective energy procurement strategies and maintenance and operation systems.
- The utility would expect high investment costs and inefficiencies as a result of soft budget limitations, such as government financial assistance or the ability to enable higher tariffs.

The government has acknowledged the need for a change in policy and has placed special emphasis on renewable energy to form a significant element of the electricity mix.

Negotiations among the key actors in the energy sector resulted in an agreement that the national utility would share with private power producers who would engage in a bidding procedure. Eskom oversaw attempts to incorporate independent power producers (IPPs) into the industry; however, these efforts were unsuccessful under the single-buyer regime. Moving forward, the key component of the power mix is to include renewables in all energy policy formulation [8,9].

This research paper addresses a major technical and economic issue by using HOMER Pro 3.18.1 software to model and improve renewable energy systems. This intends to provide a realistic and cost-effective alternative for low-income homes while also contributing to the grid's general sustainability. The study of renewable energy integration for underserved communities is also consistent with larger efforts to promote energy fairness and environmental justice.

2. Hybridization of Renewable Energies with Global Production

A power system known as hybrid/embedded technology is created by joining distributed generation technologies directly to the distribution network. A dependable and high-quality power supply for crucial loads and delicate industrial equipment is provided by integrated generation technology. This small-to-medium-sized power generation source is closed to the load points and directly connected to the electrical distribution system. Peak shaving operation, voltage management, lowering transmission and distribution line losses, and enhancing the efficiency of the suggested power system are all possible with the components of embedded power operation.

Power generation in a hybrid renewable energy system includes solar PV, wind, biomass, hydro, fuel cells, generators, and grid etc. [10]. Each of these systems' components has unique advantages and disadvantages [11]. The availability of renewable energy resources is primarily determined by factors such as resource availability, duration, meteorological conditions, and geographic location, in Figure 3 an African annual direct radiation is depicted. Because of the intermittent variance in energy, the output from wind turbines and solar photovoltaics fluctuates naturally [12,13]. In off-grid mode, the system requires an additional storage device to ensure a consistent supply of electricity to load sites. The advantage of a hybrid renewable energy systems (HRES) is that it allows power sources to compensate for each other when one is operating at a lower level. If renewable power output falls short, backup resources can be utilized to cover the remaining load. The HRES also offers a continuous supply of electricity to the load areas, with the additional power being transferred to the grid systems [14]. Integration of renewable energy systems (RESs) and conventional fossil-fuel-based generators results in hybrid energy systems (HESs) capable of overcoming the issue of intermittency and changing RES quantities. HESs can deliver more sustainable, dependable, and cost-effective systems than individual energy sources [15]. One of the most significant aspects of HRES implementation is the optimal planning and design of its components [16]. On the other hand, with correct design and planning, an on-grid connected hybrid power system will need less storage capacity, lowering the system's cost [17]. Eleven hybrid power projects signed a 20-year power purchase agreement and Eskom to supply clean power to about homes in South Africa. These projects were awarded via South Africa's Risk Mitigation Independent Power Producers Procurement Programme (RMIPPPP). The purpose of the Independent Power Producer Procurement Program (RMIPPPP) is to acquire 2000 MW of additional production capacity from various dispatchable electrical power-producing projects. The objectives of this technology-neutral program are to close the supply gap, ease long-term constraints on the supply of electricity, and lessen the widespread use of diesel-powered peaking electrical generators Figures 4 and 5 illustrate the distribution of renewable energy resources across various regions in South Africa, as well as the levels of global horizontal irradiation within the country. The data presented highlights the significant presence of solar energy as the predominant renewable resource in the country.

2.1. Solar Energy

Solar power is the process of converting solar insolation into electricity. This is achieved directly by converting the global horizontal insolation (GHI) into energy by converting photovoltaic (PV) or concentrating solar power (CSP) systems. CSP systems can also be used to achieve indirect conversion. Without any direct impact on the environment, the PV system is one of cleanness and environment friendly technology to produce power. The difference between photovoltaic and concentrated solar power methods is that photovoltaic converts solar energy into electrical energy using semiconducting materials such as silicon. In contrast, concentrated solar power methods use mirrors to generate thermal energy, which can then be used to generate electricity [18]. Thermal energy produces steam, which is then used to generate electricity [19]. South Africa is in a good position to take advantage of its plentiful solar resources because it is a developing country with an increasing power deficit issue [19,20]. The country promised to raise the share of renewable energy in the country's energy supply in a 2002 government policy paper [21]. In Figure 5, the Global Horizontal Irradiation of South Africa is shown. Figure 6 depicts the installed capacity of solar energy rooftop.

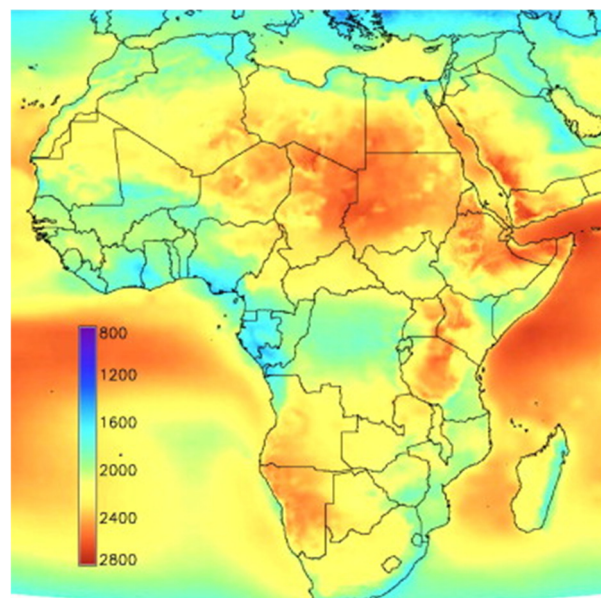


Figure 3. African Annual Direct Normal Radiation in kWh/m² [22,23].



Figure 4. Regional distribution of renewable energy resources.

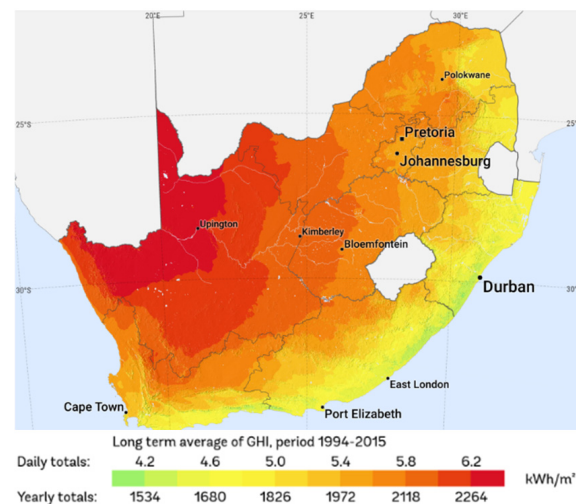


Figure 5. Global Horizontal Irradiation South Africa.

2.2. Wind Energy

The observation of the industry has revealed that the size of turbines manufactured has increased overall, and that this increase has been correlated with the higher wind energy they have been found to capture. In general, South Africa has fair to reasonable wind potential by international standards, particularly along the coastal areas of KwaZulu-Natal, the Northern Cape, the Western Cape, and the Eastern Cape. The country did not wait for the industry to develop, and it is now among the top 15 countries in Africa with the best wind resources [7]. Numerous sources have indicated that South Africa's wind resource quality and potential for producing wind power are either extremely low or of world-class level. The unique geographical circumstances, obstacles, and variations in the land surface have a significant impact on the quality of South Africa's wind resources. The potential wind market appears to be competitive with the top five wind markets globally (China, the USA, Germany, Spain, and India), based on the meso-scale wind model of South Africa that was established in 2008 [24]. Currently operating wind farms in South Africa are shown in Figure 7.

2.3. Energy Mix

The application of renewable energy technology is growing in South Africa. South Africa is considered as a country with the greatest potential for the use of renewable energy due to its climate. Figure 8 [25]. RMIPPPP projects located in 4 of the 9 provinces of South Africa are shown. Individuals, developers, and local governments are increasingly incorporating the usage of renewable technology into their development plans as they become more inexpensive [26]. Municipality-initiated renewable energy programs encounter several general obstacles and constraints that prevent their widespread implementation in South Africa. According to [27], the problems can be categorized as follows: political and legislative barriers, financial and economic issues, research and development barriers, and natural variables. Therefore, to guarantee the sustainable use of renewable energy, creative financial initiatives, appropriate planning, and political leadership are needed to address these issues. Some researchers believe that local energy end-users should be involved in energy planning. Their understanding of energy concerns as end-users, along with their knowledge of local conditions, should be completely integrated into the decision-making process. Examples of these that are more localized or integrated into the community could include the use of locally produced energy, as in the case of a local district heating structure, a "private wire" mini-grid power network, or production technologies like solar photovoltaic panels installed on community buildings. Other wind and solar technologies, wave energy, and tidal energy converters are still mostly at the pre-commercial stage, even if wind and solar PV technologies are becoming more and more widely available and

commercially developed. As a result, the possible advantages of this kind of technology vary greatly. The utilization of wind and solar technologies has the potential to significantly increase the capacity of producing electricity locally in a cleaner manner.

2.4. Renewable Energy Systems in KZ

The availability of renewable energy sources varies by area. According to the provincial assessment, solar wind and wave energy sources are abundantly available in South Africa [28]. In comparison to Europe (4.5 to 6.5 kWh/m² solar irradiance). Figure 3 shows the annual direct normal energy in Africa in kWh/m²/day, there are twice as many hours of sunlight each year [29]. As a result, solar energy is available via photovoltaic (PV) panels or thermal technologies such as concentrating solar panels (CSP) (547.6 GW), Free-State (25 GW), and wind power (1 GW).

South Africa's renewable resource potential is substantial, but it has yet to be completely realized and used to benefit the energy sector, society, and the economy. In November 2003, the country announced its first renewable energy (RE) policy document, a White Paper. This vision was formalized in the Integrated Resource Plan for Electricity (IRP 2010–2030), which was gazetted on 6 May 2011 [7], KZN makes a large contribution to solar energy in South Africa due to its constant high temperature [30,31]. Western Cape has effectively integrated solar energy with contempt, reflecting KZN's inability to sustain solar electricity, resulting in the lack of sponsors for any solar project in the province. the resource mapping, the type and magnitude of RE opportunities varies by province, Figure 4 depicts the most prominent RE opportunities per province. The RE resources not indicated are not non-existent, but they are minor in comparison to the highlighted resource options. For example, landfill and cogeneration options exist in many major cities and are not explicitly shown here.

KZN receives more than enough sunlight to power its solar energy systems. This paper focuses on the economic uncertainties associated with the integration of renewable energy with the current grid supply, which will help meet the region's energy demands at the lowest cost based on the mostly available sources such as solar PV and wind technologies.

2.5. Energy Prices and Energy Demand

The low-income households in rural areas use freely available sources of energy like cow dung and firewood to save money [32]. Despite the availability of many energy technologies in South Africa, rural households continue to struggle to meet their energy needs. For several years, certain rural regions have struggled with insufficient national grid coverage and poor electrical supplies. The government has made numerous efforts to establish grid connections in South Africa's KwaZulu Natal province. However, even in electrified areas, people are unable to pay due to poverty, demonstrating that energy poverty renders access to power unpredictable. Rural areas in KwaZulu Natal are among the South African regions with the lowest access to renewable energy, despite abundant energy resources such as solar, biomass, and micro-hydrokinetic energy.

The high investment costs of solar water heating technology (SWH), along with low income levels in South Africa's rural areas, have made large-scale deployment of solar water heating systems a difficult job. On the one hand, solar technology systems have a relatively high investment cost but a lower maintenance cost; on the other hand, fossil fuel-based heaters have a lower initial investment cost but a higher operating cost for users. This implies that investors without considerable cash flow will be deterred from investing in the SWH firm [33]. ESKOM's subsidy effort, which subsidized SWHs, allowed for exponential growth of the business prior to its expiration in 2015; nonetheless, this discontinuation has resulted in slow growth in the industry [34], which could have been avoided if new subsidies or comparable initiatives were implemented. Although the SWH business in South Africa is developing, even if slowly, the government may do more to provide incentives to investors and customers, allowing the industry to attain its full potential.

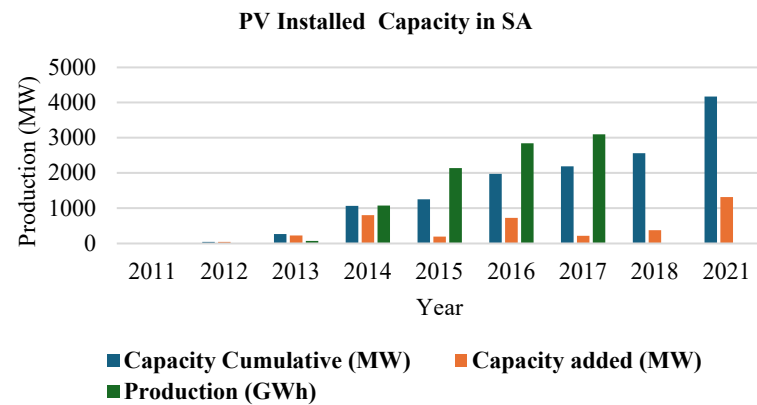


Figure 6. Rooftop PV installed capacity for low-income households [35].

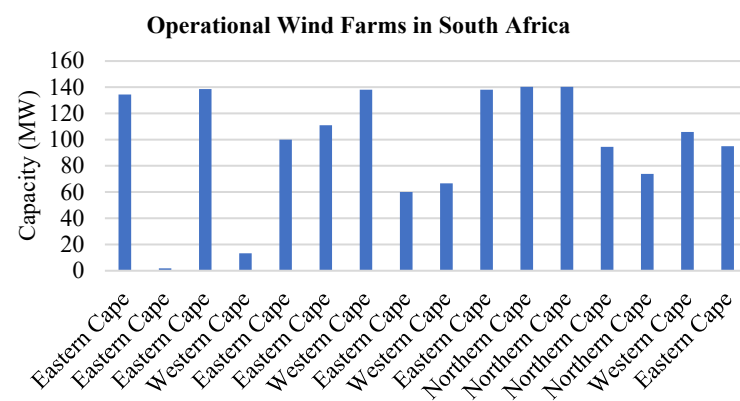


Figure 7. Currently operating wind farms in South Africa [36].

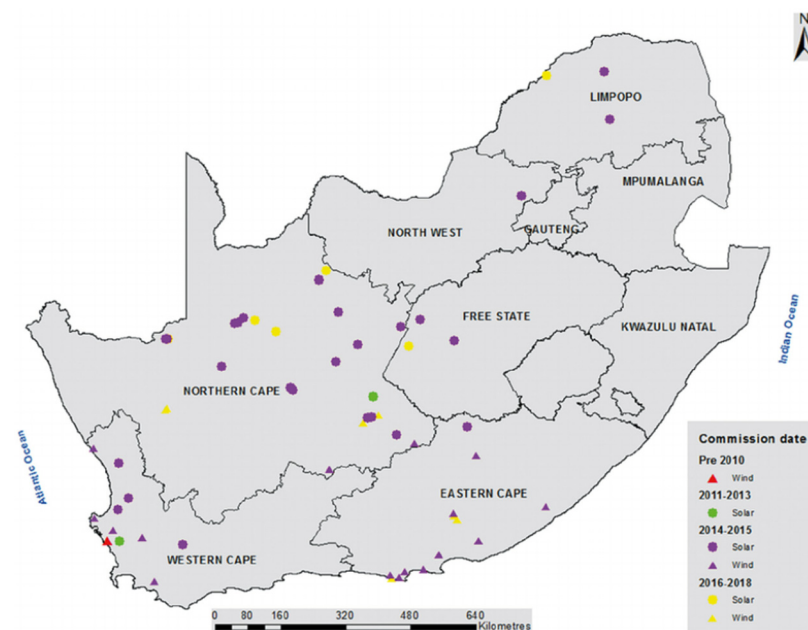


Figure 8. The locations of each province's RIPPPP projects.

3. Technical Modeling of a Hybrid System

This study offers the optimal hybrid energy system architecture to meet the load requirements of a particular on-grid residential. With one of the main objectives requiring an understanding of an appropriate model for the selection of renewable energy technologies in South Africa, it was necessary to utilize a methodology that allows for the selection of re-

newable energy technologies connected to the current grid in an effective manner. Figure 9 shows the hybrid energy system considered for this study consisting of the PV system and the wind system. The output of the PV arrays is integrated into a DC bus and the wind turbine into an AC bus. The eThekweni municipality has not finalized its small-scale embedded generating systems' payback timelines and price reduction to energy users, even though it is usually the case for a grid-connected hybrid system to inject its excess generated power back into the grid. As a result, this study investigates payback periods and attempts to determine the most economical energy mix solution. Using HOMER Pro 3.18.1 software, costs were calculated by balancing hybrid energy systems and their associated components with current constraints, load profiles, and meteorological data. The HOMER Pro software was utilized to compute the net present cost (NPC) and cost of energy (COE). This cost computation served as the basis for the optimal system's comparison study. The optimization and sensitivity analysis features of the HOMER Pro program enable the technological and economic feasibility of hybrid energy system configurations to be examined for on-grid applications. Figure 10 shows the process of the HOMER Pro software.

3.1. Selected Site Location

The worksite chosen is situated on the South Coast of Durban, KwaZulu Natal. The area's current load demand is entirely dependent on grid supply, with annual operating energy expenses amounting to R1.06 million. However, the increasing load demand in the selected region is resulting in an electricity shortage. To address this issue, additional power or a grid extension must be installed. Extensive research is required to find a solution, leading to the integration of renewable energy sources into the existing grid to meet the growing demands. Figure 10 depicts the proposed grid hybrid scheme, which is intended to ensure electrical power demand in rural KwaZulu Natal. On average, each resident consumes 11.13 kWh daily, with a peak load of 2.07 kW. Figure 11 shows the schematic on-grid system configuration for the study analysis. In Figure 12, a load profile for the study area generated by the HOMER Pro software is depicted while Figure 13 represents a consistent high solar irradiation throughout the year, with notable peaks in January, February, November, and December. During these months, the measured irradiation levels were 5.31, 5.21, 5.02, and 5.02 kWh/m²/day. The graph shows a gradual decrease in solar radiation from January to June, with the lowest point in June, followed by a gradual increase from July to December, when it reaches its peak. Solar irradiation varies according to the season, with more direct sunlight in the summer when the sun is directly overhead and less direct sunlight in the winter. In Figure 14, the wind speed data predicted from the prediction of worldwide energy resource database are represented. Furthermore, other factors such as, air pollution, cloud cover, etc. can all reduce the intensity of solar radiation.

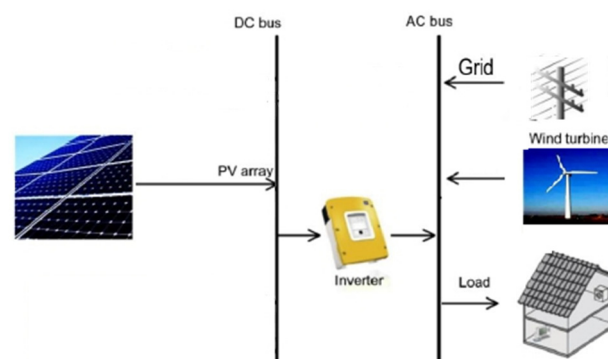


Figure 9. Diagram of a hybrid renewable energy system connected to the grid.

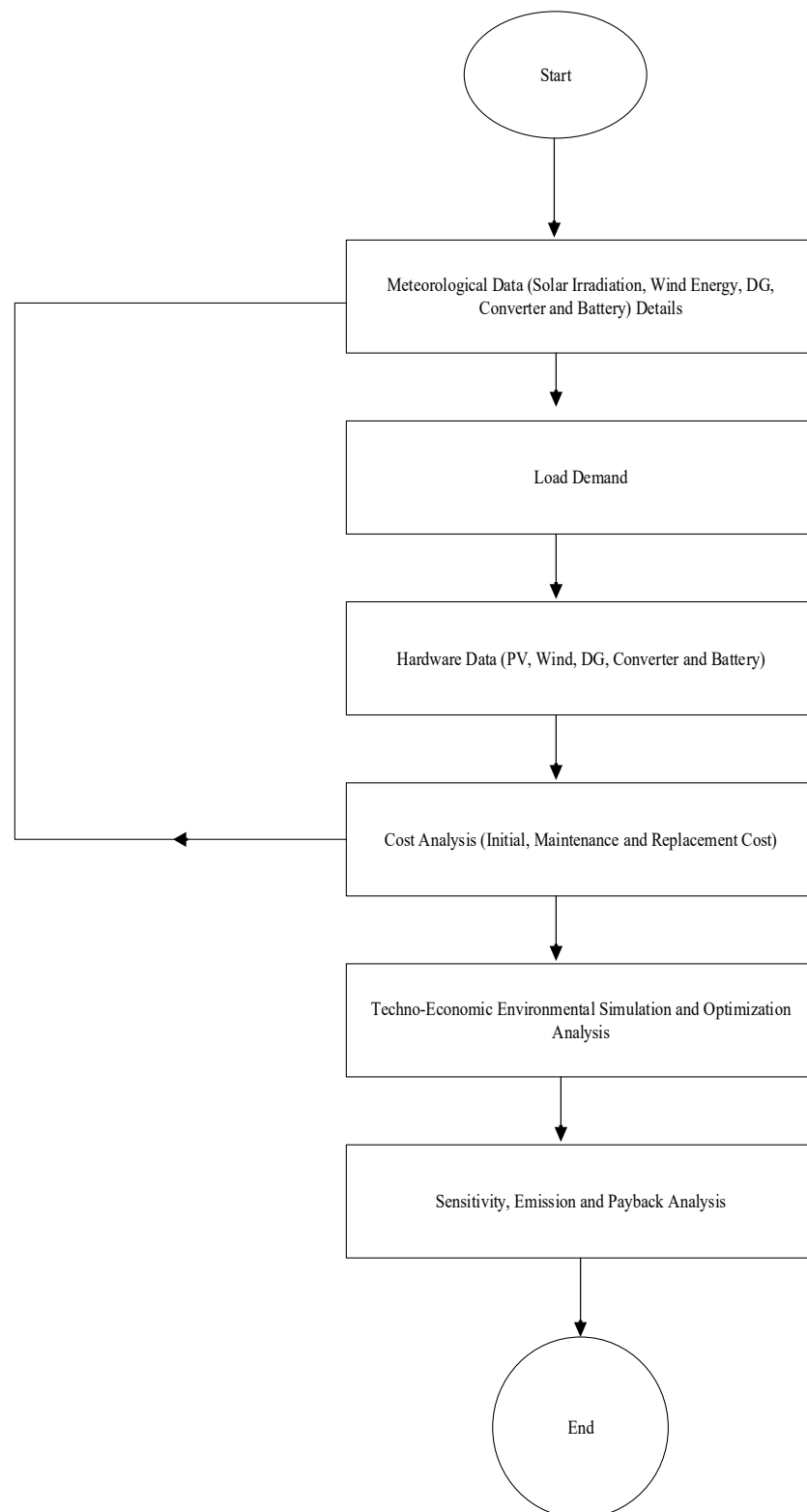


Figure 10. Proposed methodology of the hybrid renewable energy system.

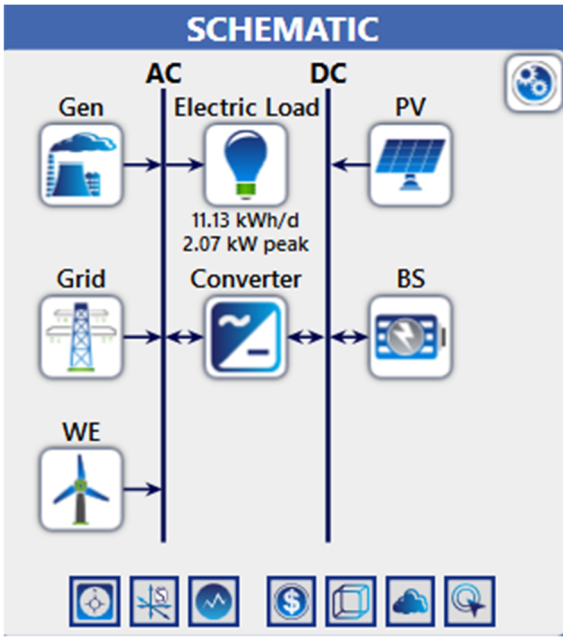


Figure 11. On-grid system schematic configuration.

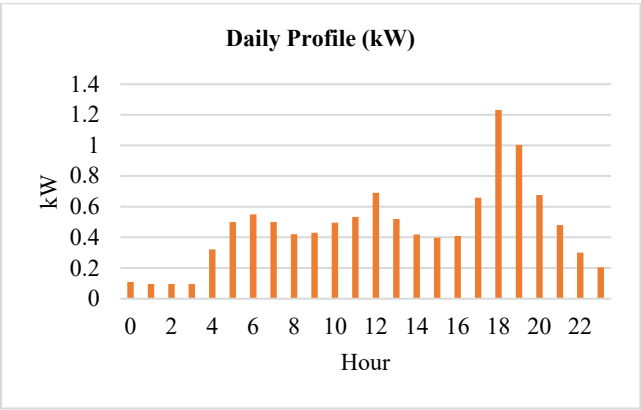


Figure 12. Daily load profile during summer.

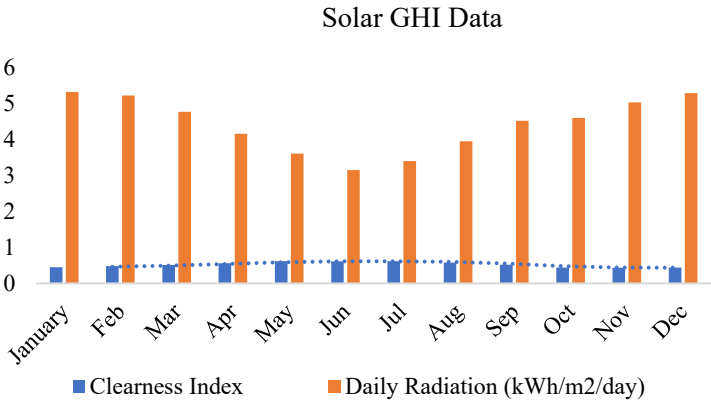


Figure 13. Monthly average solar global horizontal irradiance (GHI) data.

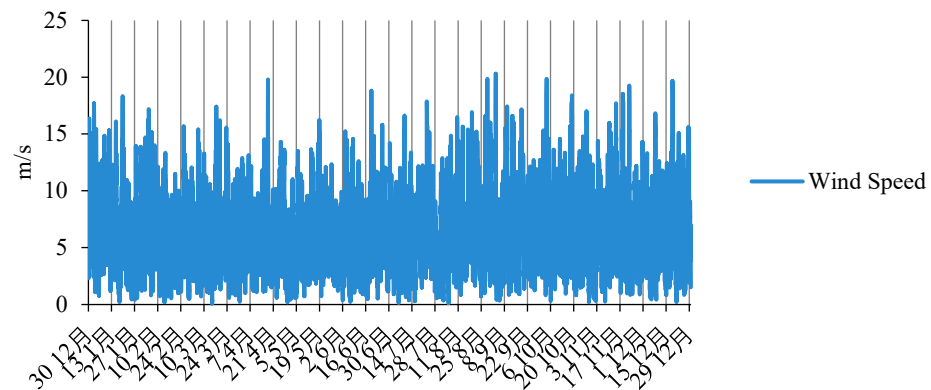


Figure 14. Monthly Average Wind speed data.

3.2. Solar PV

PV modules use sunlight and ambient temperature to generate DC electricity. This technology is widely used. The power generated by PV panels under specified conditions was calculated simply by HOMER using the formula shown in Equation (1), considering the temperature impact factor [37].

The temperature of PV cells contributes to their performance and efficiency; higher temperatures can reduce efficiency by increasing material resistance. Furthermore, variations in solar radiation are considered when calculating efficiency. The efficiency of the PV system varies throughout the day, depending on solar radiation and weather patterns. The primary characteristics of a PV panel in any hybrid system include efficiency based on output power generation, cost, secure layout, maintaining power production efficiency over long periods of time (the selected project lifespan), and the ability to generate power in low solar radiation scenarios.

The basic flat plate PV with a 12 kW capacity that is being studied cost ZAR 125,000 in capital and replacement costs and ZAR 3000 in operating and maintenance costs annually. A life expectancy of 25 years and an 80% de-rating factor were considered. The clearness index shows monthly average global radiation statistics from National Aeronautics and Space Administration (NASA) using geographic coordinate information. Figure 13 shows the monthly average solar global horizontal irradiance data.

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{\bar{G}_T}{\bar{G}_{T,STC}} \right) \quad (1)$$

3.3. The Converter

There are two modes of operation for a converter: rectification and inversion. The peak demand and electricity consumption are crucial considerations for choosing the right converter size. A bidirectional converter is used in this study to serve as both an inverter to supply the load demand when necessary and a rectifier to store excess power in the battery [38]. Electronic converters play an important role in balancing power transfer between AC and DC elements in hybrid renewable energy systems. These converters serve as both rectifiers and inverters, enabling energy management by switching between different types of electrical power generation (rectifying mode AC to DC and inverting mode DC to AC) while maintaining the required frequency for load demands. When there are no solar or wind resources available, the converter operates solely in inverter mode; this condition is most common at night and during cloudy weather. When there is an adequate amount of renewable energy available, the converter works in rectifier mode only and charges the battery storage system.

In this study, a general system converter with 95% efficiency is taken into consideration. The formula of the converter efficiency is depicted in Equation (2). For every 12 kW, the

capital expenditure and overall cost of replacing the converter is ZAR 125,000. With a 15-year lifespan, the annual maintenance and operating costs is to ZAR 3000 [39].

$$\eta_{\text{conv}} = \frac{P_{\text{out}}}{P_{\text{in}}} \quad (2)$$

3.4. Battery Modeling

Energy storage is one of the main elements of the hybrid setup. It makes a significant contribution to improving system reliability. When there is a scarcity of energy, the battery is utilized to store the excess electricity produced by other power sources and release it [40]. When AC power is converted to DC power, stored in a battery, and then converted back to AC power to meet the demand for electricity, the relationship between the total system efficiency and the battery round-trip efficiency is shown in Equation (3) [39,41]. Round-trip efficiency in batteries is defined as the ratio of useful output energy to useful input energy. A generic battery of rating 12 V, 1 kWh lead-acid battery of energy storage is considered for this study. The battery's capital cost and replacement cost are ZAR 50,000 respectively. The lifetime of the battery is 10 years with the throughput of 800 kWh. Energy generated by the system is collected for battery storage and provided to the load during an electrical outage.

$$\eta_{\text{overall}} = \eta_{\text{inv}} \eta_{\text{rect}} \eta_{\text{rt}} \quad (3)$$

3.5. Wind Energy

Wind energy has grown significantly over the last decade, and it is now the world's second largest renewable source of electricity generation. Wind turbines (WTs) are commonly used to generate electrical power from wind energy via an appropriate wind energy conversion system. Turbines convert wind energy into electrical energy. The selection of wind turbines is influenced by various factors such as hub height, component cost, service time, and cut-in wind speed. Because of this, the power output that can be obtained from various wind generators differs significantly and depends on the wind speed at hub height. HOMER Pro determines the wind speed at the wind turbine's hub height at each time step. The equation shown in 4 is used by HOMER to determine the hub height wind speed with the logarithmic law. The power generated by WT is primarily determined by wind speed characteristics at the location of installation and WT specifications, as shown in Equation (5) [42].

The wind resources information was extracted from the solar energy database and the NASA meteorology HOMER website for the selected location. The scaled average wind speed of the selected area is 6.14 m/s, with a hub height of 17 m and an expected lifespan of 15 years.

$$U_{\text{hub}} = U_{\text{anem}} \cdot \frac{\ln(Z_{\text{hub}}/Z_0)}{\ln(Z_{\text{anem}}/Z_0)} \quad (4)$$

$$P_{WT} = \begin{cases} P_r^{WT} \times \frac{(V - V_{\text{in}})^3}{(V_r - V_{\text{in}})^3} & V_{\text{cin}} \leq V \leq V_r \\ P_r^{WT} & V_r \leq V \leq V_{\text{cin}} \\ 0 & V \geq V_{\text{out}} \leq V_{\text{cin}} \end{cases} \quad (5)$$

3.6. The Grid

One of the most important components of the electrical power system is the grid, which is the current system supply for the selected area. When renewable energy sources are unable to provide electricity at night, it is possible to purchase electricity from the grid. Similarly, when renewable energy sources produce more energy during the day, it is sold to the grid and the bill is settled through net metering. In technical terms, these are called sell-back rates, demand rates, and grid pricing. For example, when solar PV generates excess electricity during the day, it can be sold back to the grid. Similarly, in the absence of renewable energy production and a charged battery, electricity would be purchased from the grid if available.

The net electrical energy supplied to (by) the grid is the difference between electrical energy purchased and electrical energy sold to the grid, as shown in Equation (6) [43].

$$\text{GEC} = \sum_p^r \sum_q^{12} \begin{cases} E_{j.net(G),p,q}^{ecG,p} & \text{if } E_{j.net(G),p,q} \geq 0 \\ E_{j.net(G),p,q}^{ecS,p} & \text{if } E_{j.net(G),p,q} < 0 \end{cases} \quad (6)$$

In the HOMER model, the grid is viewed as a component that the micropower system can buy and sell AC electricity from. An energy charge based on the quantity of energy purchased in a billing period plus a demand fee based on the peak demand throughout the billing period might make up the cost of buying power from the grid.

3.7. Diesel Generator

A diesel generator serves as a backup power source in hybrid systems to ensure continuous power supply. In the absence of renewable resources and grid power, a diesel generator will be used to meet electrical power demand reliably. The annual fuel (diesel) consumption value (FC) is calculated using the Equation (7) [44].

Users of HOMER can add to the library of various predetermined fuels that the program offers if needed. A fuel's density, lower heating value, carbon and sulfur contents are among its physical characteristics. Additionally, the user has the option of selecting L, m³, or kg as the most suitable measurement units. The price of the gasoline and any applicable annual use limit are its two remaining characteristics [45]. Diesel cost is required to propose or model any power system. The South African Petroleum Corporation reports that the current diesel price in South Africa is ZAR 23.9 per liter. The price has changed drastically over time. Because diesel fuel prices are strongly correlated with NPC and COE for power output per unit, any diesel-based microgrid project may be delayed or even cancelled as a result of this volatile behavior [46].

$$F_C = F_1 R_D + F_2 P_D \quad (7)$$

4. Economical Modeling of a Hybrid System

This software is suitable for designing hybrid renewable energy systems, including off-grid and grid-connected systems, and performing techno-economic analyses. Different component sizes and combinations are entered as input data, along with the defined load, and a simulation is performed to produce the optimal system in terms of NPC, COE, and operating cost.

The NPC covers the price of starting a project, replacing parts, maintaining it, and fueling it. It also covers the cost of purchasing electricity from the grid and other expenses, like fines for emitting pollutants. Sales of electricity to the grid and salvage value realized at the end of the project's lifespan are included in revenues. The initial capital cost, which happens in year zero, replacement cost, which happens every time a component needs to be replaced at the end of its lifetime, and operation and maintenance cost, which happens every year of the project lifetime, for each component of the system were specified in the study.

4.1. Net Present Cost

The HOMER's simulation and optimization processes, which search for the system configuration with the lowest total net present cost, depend heavily on economics. In the simulation phase, the system is operated to minimize the total net present cost. The HOMER optimization process compares the economics of several system topologies with different proportions of renewable and non-renewable energy sources. Such comparisons need to take operating and capital costs into consideration to be fair. To do this, life-cycle cost analysis incorporates all expenditures incurred during the system's lifetime. A system's life-cycle cost is represented by the total net present cost (NPC) in HOMER

The total net cost of the hybrid system is the present cost of all costs incurred by the system during its defined useful life, minus the salvage value over that time. The net

present cost includes capital, replacement, operation, and maintenance costs, as defined by Equation (8). Homer Pro software computes the NPC of each component in the presented system. To determine the annualized cost, the homer first determines the net present cost and then multiplies it by the capital recovery factor, as shown in Equation (9) below:

$$C_{NPC} = \frac{C_{ann,tot}}{CRF(i, R_{proj})} \quad (8)$$

where,

$$\begin{aligned} CRF(i, N) &= \frac{i(1+i)^N}{(1+i)^N - 1} \\ BP &= \frac{\sum_{i=1}^n ICC_i}{AOC} \\ COE &= \frac{C_{ann,tot}}{E_{prim} + E_{def} + E_{grid,sales}} \end{aligned} \quad (9)$$

4.2. Grid Energy Charge and Demand Costs

HOMER considers the grid as a component that allows the micropower system to purchase and sell alternating current electricity. Purchasing power from the grid involves two charges: an energy charge for energy purchased and a demand charge for peak demand during the billing period. HOMER defines the grid power price as the price that the electric utility charges for energy purchased from the grid, and the demand rate as the price (dollars per kilowatt per month) that the utility charges for peak grid demand. The sellback rate is the price the utility pays for power sold to the grid. Because the grid power price fluctuates hourly as the applicable rate changes, the grid's marginal cost of energy does as well. This could have a significant impact on HOMER's simulation of the system's behavior. For example, HOMER may choose to run a generator only during periods of high grid power prices, when the cost of grid power exceeds the cost of generator power. Equation (10) shows the total annual energy charge.

$$GC_{grid,energy} = \sum_k \sum_j^{rates\ 12} E_{gp,k,j} \times C_{pr,k} - \sum_k \sum_j^{rates\ 12} E_{gs,k,j} \times C_{cb,k} \quad (10)$$

4.3. Payback Period

The HOMER Pro uses system comparisons to determine payback. Payback often indicates the number of years needed for the return of an investment. The payback period is the number of years it takes for the total revenue from an original investment to match the initial investment value. Independent power providers and microgrid operators can evaluate a project's short- and long-term benefits by looking at its payback period, which serves as a crucial performance measure. Payback only makes sense when one system is contracted with another. By contrasting the difference in capital and operational costs, the payback of the additional investment needed for the PV–diesel–battery combination can be determined. That is precisely what HOMER Pro's Compare Economics window accomplishes [47]. A microgrid system that has a shorter payback period is economically viable and operates with higher sustainability when compared to a project that has a longer payback period, as shown in Equation (11) [48].

$$ROI = \sum_{i=0}^{R_{proj}} \frac{C_{i,br} - C_i}{R_{proj}(C_c - C_{c,ref})} \quad (11)$$

5. Results

5.1. Optimization Results

A variety of system configurations eliminated the infeasible ones, ranked the feasible ones according to total net present cost, and selected the feasible one with the lowest total net present cost as the optimal system configuration. The optimization process identified

the best possible, or optimal, system configuration that satisfied the constraints at the lowest total net present cost as shown in Table 1. When integrating a 90 kW wind generation capacity into the current grid system, the proposed configuration proved to be the most efficient among all alternatives. It exhibited the lowest net present cost (NPC) and levelized cost of energy (LCO) while also minimizing energy purchases from the grid and maximizing energy sales to the grid, as shown in Table 1. The power generation from this optimized, cost-effective system is depicted in Figure 15. The alternative that ranked just below the top choice involved integrating photovoltaic (PV) solar energy with the grid, which exhibited a lower net present cost (NPC) and levelized cost of energy (LCO) compared to wind energy. However, the energy pricing differed slightly from the existing system, and the amount purchased from the grid was higher in comparison to the wind energy/grid configuration. The decision regarding the most appropriate design will ultimately rest with the designer, who must consider the specific site and design constraints. This situation results in a minimal amount of energy being sold back to the grid, necessitating a thorough investigation to address the issue.

Table 1. Optimization results for system configurations.

System Configuration	PV Production (kWh/yr)	Wind Energy Production (kWh/yr)	Grid (kW)	NPC (R)	LCOE (R/kWh)	Operating Cost (R/kWh)	RF (%)	IRR (%)	Simple Payback (yr)
WE/Grid	-	190,725	999.999	−2.38M	−0.628	−188,485	99.7	15	6.3
PV/WE/Grid/CONV	1253	190,725	999.999	−2.36M	−0.624	−188,411	99.7	15	6.3
WE/DG/Grid	-	190,725	999.999	−2.33M	−0.617	−192,036	99.7	14	6.8
PV/WE/DG/Grid/CONV	1253	190,725	999.999	−2.32M	−0.612	−191,963	99.7	13	6.9
WE/BT/Grid/CONV	-	190,725	999.999	−2.27M	−0.599	−185,428	99.7	14	6.5
PV/WE/BT/Grid/CONV	1253	190,725	999.999	−2.25M	−0.595	−185,358	99.7	14	6.6
WE/DG/BT/Grid/CONV	-	190,725	999.999	−2.22M	−0.587	−18,898	99.7	13	7
PV/WE/DG/BT/Grid /CONV	1253	190,725	999.999	−2.21M	−0.583	−188,910	99.7	13	7.1
PV/Grid/CONV	1253	-		237,995	2.96	10,153	19.1	21	20
Grid	-	-		238,721	2.97	12,065	0	0	0
PV/DG/Grid/CONV	1253	-		282,728	3.52	6602	19.1	0.24	25
DG/Grid	-	-		283,453	3.53	8514	0	0	0
PV/BT/Grid/CONV	1253	-		348,413	4.33	13,174	19.4	0	0
BT/Grid/CONV	-	-		349,849	4.35	15,122	0	0	0
PV/DG/BT/Grid/CONV	1253	-		393,146	4.89	9622	19.4	0	0
DG/BT/Grid/CONV	-	-		394,581	4.91	11,571	0	0	0

WE(wind energy),PV (photovoltaic), CONV (Converter), BT(battery/batteries) and DG(diesel generator).

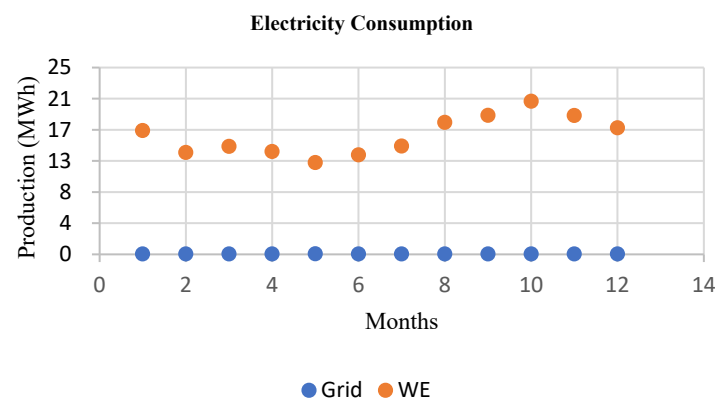


Figure 15. Electricity production for the require demand.

In this system, the amount of energy purchased from the grid is very low, as shown in Figure 16. This results in more excessive power produced by the wind, which is sold to the grid which resulted in the less amount of power purchased from the grid. Table 2

shows the systems electrical outputs. The findings indicate that the proposed system can meet the demand load with an annual electricity production of approximately 191,360 kWh. When the maximum renewable percentage reaches 100%, the renewable fraction stands at 99.7%. Consequently, the surplus power sold to the grid amounts to 187,297 kWh per year, representing 98% of the electricity sold to the grid derived from renewable energy sources. This achievement significantly decreases the energy purchased from the grid and lowers the overall electricity costs. The cost summary for the optimized system is displayed in Figure 17, while Figure 18 provides a depiction of a cash flow analysis.

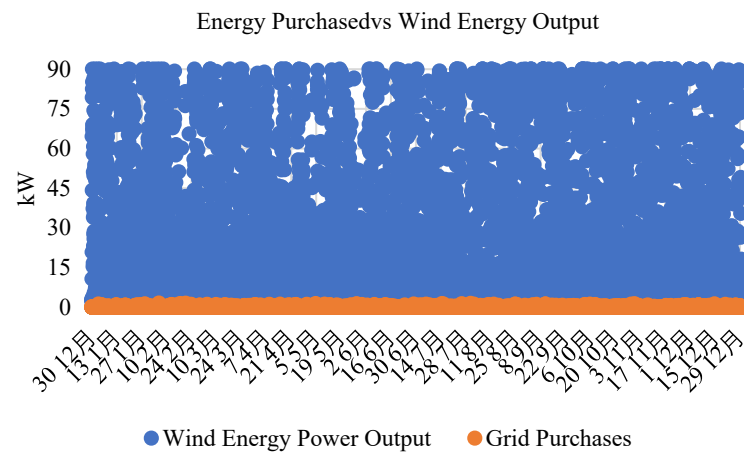


Figure 16. Energy purchased from the grid from wind power production.

Table 2. Grid electrical system architecture results.

Electrical	Component	kWh/y
Production	Grid Purchases	635
	Wind Production	190,725
Consumption	AC Primary Load	4062
Quantity	Excess Electricity	0
	Unmet Load	0
	Capacity shortage	0
	Renewable Fraction	99.7%

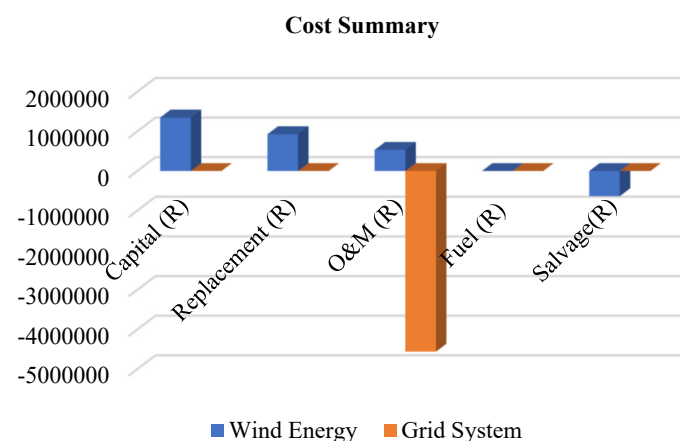


Figure 17. Cost summary of the proposed system configuration.

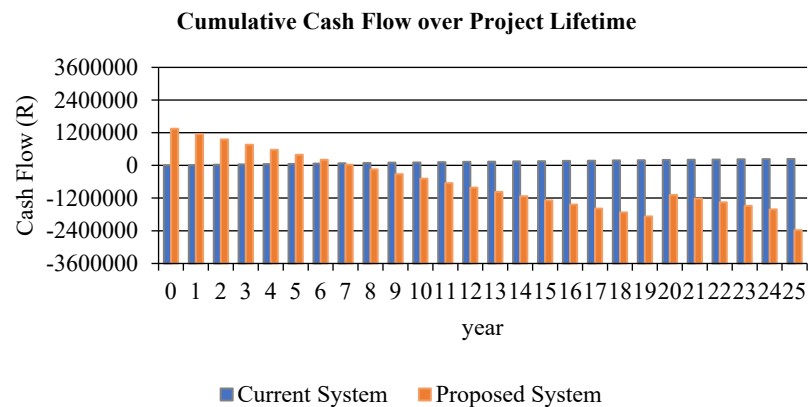


Figure 18. Cumulative cash analysis over the project's lifetime.

This study uses NPC and COE for techno-economic evaluations and feasibility studies to identify multiple ideal configurations. Sensitive variables such as a fuel price of ZAR 23.9/L, a wind speed of 6.14 m/s, and solar radiation of 4.41 kWh/m²/d as annual averages are considered in the optimization findings. The economic evaluation of hybrid generation systems utilized a HOMER cycle charging dispatch strategy to simulate various scenarios. The hybrid system with the lowest NPC, consisting of 4.56 kW of PV and 1.31 kW of converter, was identified as the optimal solution. This system, integrated with the national grid, has an NPC of ZAR 235,847, while the base case system's NPC is ZAR 301,637. Operating costs are reduced to R2064/year with a payback period of 19 years and an IRR of 2.49%. The scenarios are ranked based on NPC value, from highest to lowest.

5.2. Sensitivity Analysis

This study conducted a sensitivity analysis of on-grid systems integrated with renewable energy sources, focusing on the variation in parameters such as grid price (R/kWh), net present cost (NPC), operations and maintenance (O&M), cost of energy (COEs), and renewable fraction (RF). The sensitivity parameters chosen for this analysis included global solar irradiation, annual wind speed, and fuel price. For each variable, the base case values were adjusted either upwards or downwards. The actual base case values for global solar irradiation, annual wind speed, and fuel price were determined to be 4.41 kWh/m²/day, 6.14 m/s, and 23.9 R/L, respectively. The sensitivity results reveal that with global solar radiation at 3.9 kWh/m²/day, a wind speed of 5.83 m/s, and a diesel price of 23.9 ZAR/L, there is a decrease in the net present cost (NPC), cost of energy (COE), and operational and maintenance (O&M) costs. Nevertheless, the power price remains consistent with the current/base system. The renewable factor is lower when compared to the optimal system. Figure 19 shows the sensitivity result for the optimal grid system and net present cost analysis, respectively. As a result, the energy purchased from the grid increases to 635 kWh, while the grid price stays the same at ZAR 2.97/kWh, and the energy sold to the grid drops to 190,725 kWh annually. The simple payback period for this setup is longer, at 6.3 years. In contrast, when the fuel price escalates to ZAR 25/L, with a wind speed of 6.5 m/s and solar energy at 4.59 kWh/m²/day, the grid price rises to 3 ZAR/kWh. This scenario results in a higher energy production of 243,486 kWh per year, leading to a decrease in energy purchased from the grid to 503 kWh and an increase in energy sold back to the grid. The payback time period or the scenario is reduced to 4.3 years, which is depicted in Figure 20, while Figure 21 represents power production for power generation sources. The findings suggest that further investigation is warranted regarding the integration of renewable energy sources into the grid, particularly concerning the sellback time.

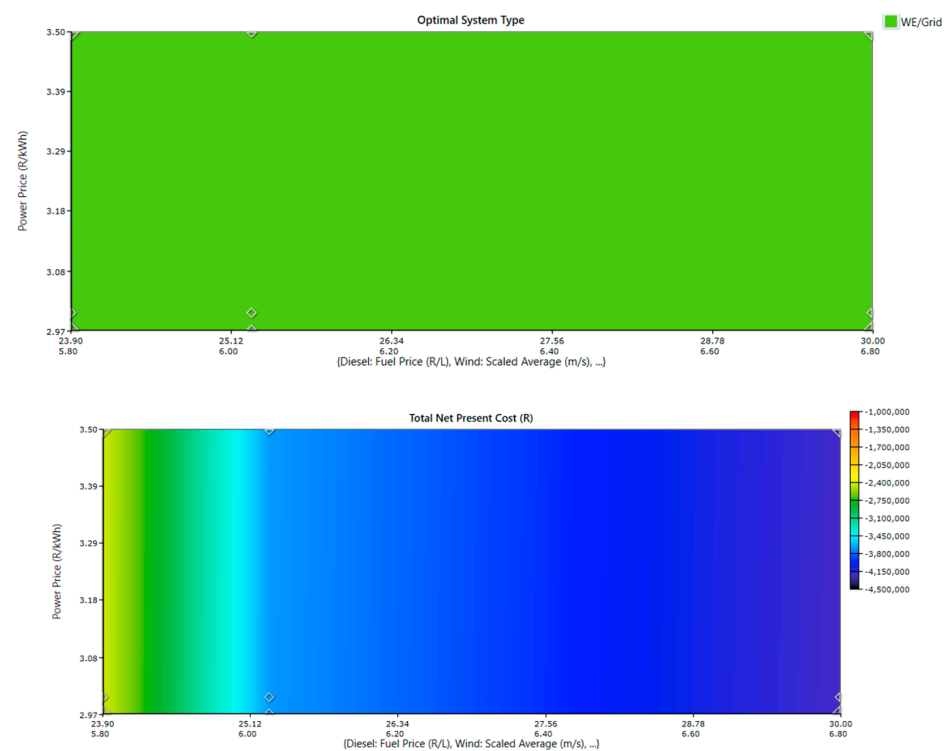


Figure 19. The repercussions of the simultaneous variation in solar radiation, wind speed, and fuel pricing on the most efficient system type for cost of energy (COE).

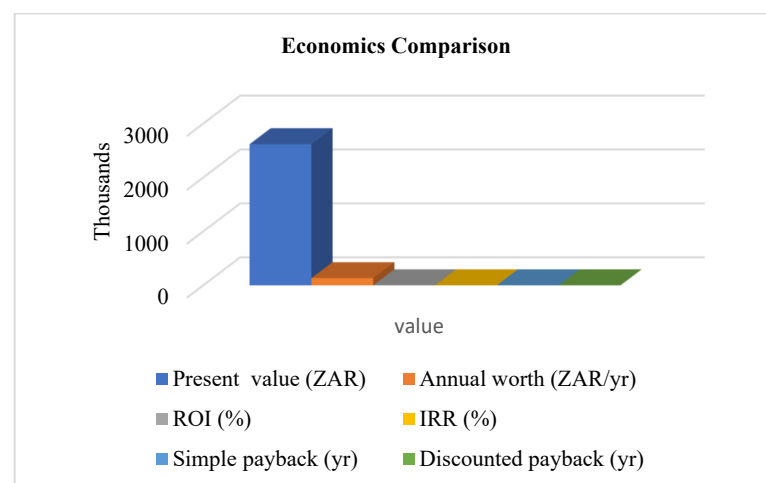


Figure 20. Economic comparison of the base and the proposed system.

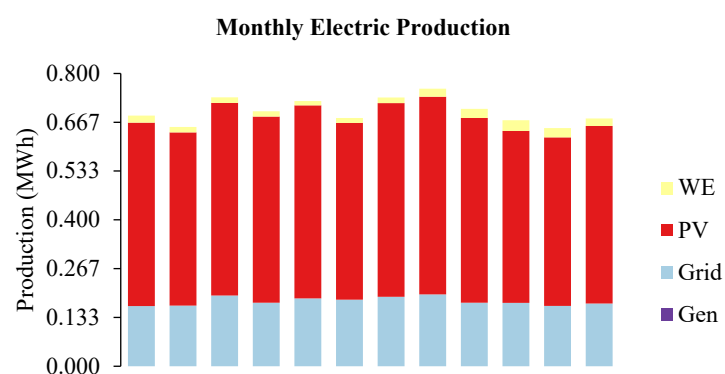


Figure 21. PV/Wind/DG/Grid/Converter-based electricity production throughout the year.

6. Conclusions

This study investigates the feasibility of a hybrid system for projected energy generation at a designated area on the south coast of KZN. Different possibilities are assessed, and a configuration based on reduced NPC is proposed. According to HOMER, this system consists of a WE/Converter/Grid scenario. The integration of 90 kW of wind energy into the grid results in a complete elimination of energy costs. The findings indicate that the chosen location can effectively support an off-grid system, which would adequately meet the energy demands of that area. Furthermore, the analysis reveals that the amount of energy procured from the grid is significantly lower when compared to other system configurations. An alternative system worth considering is the PV/Grid option; however, there is a minor distinction in the energy costs incurred from the grid.

The NPC is ZAR 235,846.80, with a renewable percentage of 74%. The PV system generates around 6.588 kWh per year, with a grid purchase of 2213 kWh and a grid sale of 2.154 kWh. The performance of a hybrid energy system can be assessed by considering costs, power quality, benefits, etc. Results from a study of a grid-connected renewable energy hybrid system using some key performance indicators can be used to locate and measure solar and battery storage units. The cost and benefits, such as reduced environmental pollution, annual energy savings, and revenue, are important considerations for power sector stakeholders when making managerial decisions about the installation of renewable energy units in the grid power network. This ensures that integrated renewable energy sources benefit both consumers and the generation industry.

The results show that integrating renewable energy sources into a grid power network has financial benefits. The paper's findings can be used by key players, financiers, and local partners in the renewable energy sector to ensure the successful implementation of renewable energy projects. This work also presented the technical and financial indicators that many countries must consider when developing renewable energy projects. The study's findings also show that incorporating optimally sized resources such as wind into the distribution system is both economically and technically viable. These findings can help the South African government and international organizations achieve environmentally friendly electricity.

From findings, on-grid systems have lower renewable energy penetration and require no or limited storage capacity. This lowers energy costs, but it also wastes excess energy and causes instability in the network system. This energy could be used wisely, either by storing it in batteries or by powering other variable loads. As a result, additional research can be conducted to determine the proper sizing of energy storage to avoid stability issues such as power unbalance, battery damage, and energy waste from the network that can be sold back to the grid to reduce energy costs.

This will further improve the system's efficiency and reliability. Further research can be carried out on different components of effectively using excess energy from a hybrid grid-connected system, as well as considering optimal sizing allocation of battery storage systems for hybrid energy systems.

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Abbreviations

AOC	Annual operating cash flow
$C_{ann,tot}$	Total annualized cost
C_c	Capital cost of the system
$C_{c,ref}$	Capital cost for reference system
C_i	Current system Nominal annual cash flow
$C_{i,br}$	Reference system Nominal annual cash flow
$C_{cb,k}$	Sellback rate for rate k
$C_{pr,k}$	Power price of the grid
CRF	Capital recovery factor
E_{def}	Total amounts of deferrable load
$E_{gs,k,j}$	Amount of energy sold to the grid
$E_{gp,k,j}$	Total amount of energy sold to the grid
$E_{grid,sales}$	The amount of energy sold to the grid per year
$E_{j,net(G),p,q}$	Net energy purchased in kWh from the grid in month q during the applicable rate (r) of p
E_{prim}	Total amounts of primary load
F_1	Coefficient of intercept
F_2	Fuel curve slope
f_{pv}	PV derating factor
GEC	Energy charges in kWh
$g_{ecG,p}$	Power price of grid for rate p in ZAR/kWh or USD/kWh
$g_{ecS,p}$	Selling price of grid for rate p ZAR/kWh or USD/kWh
$\overline{G}_{T,STC}$	Incident radiation at standard test conditions
i	Annual real interest rate (the discount rate)
ICC	Initial capital cost
ln	Natural logarithm
N	Number of years
P_D	Diesel operated generator outcome
P_{in}	Input power of a converter
P_{out}	Output power of converter
P_r^{WT}	Rated power generated from the wind turbine
P_{WT}	Power generated from wind speed
R_{proj}	Project lifetime
U_{anem}	Wind speed at anemometer height
U_{hub}	Wind speed at the hub height of the wind turbine
V_{cin}	Cut in wind speed
V_r	Rated wind speed
V_{out}	Cut off wind speed
Z_{hub}	Hub height of the wind turbine
Z_0	Surface roughness length
η_{inv}	Inverter efficiency
η_{rect}	Rectifier efficiency
η_{rt}	Battery round trip efficiency results

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